Electric Motor Drive and Power Conversion

Student learning guide | Edward Selorm | BEV laboratory

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Understand the inputs, follow the power path and verify the results. The guide matches the on-page learning section.

Find your way around

Left: choose an experiment and set the inputs. Right: inspect calculated results, playback, the drivetrain and graphs. The opening baseline calculates automatically. Change a setting and select Calculate experiment to replace the results. Invalid settings retain the previous completed run.

Your first drive experiment

Keep the baseline preset: target 1500 rpm, initial load 10 N m, final load 20 N m, load change at 2 s. After calculation is ready, Play results moves through the stored response; Pause holds the selected instant. Drag Simulation time to inspect any point. At 5 s, motor speed should be very close to 1500 rpm. The controller supplies extra torque after the 2 s load change.

Three different meanings of time

Load change time is WHEN the external load changes: 2 means two virtual seconds after the start. The solver calculation interval is HOW OFTEN equations are updated: averaged mode uses 0.00002 s (20 microseconds), so 50,000 steps cover one virtual second. Recording interval is HOW OFTEN a row is saved: default 0.001 s (one millisecond), giving 5001 rows from 0 to 5 s inclusive. Playback speed only changes how quickly you view those stored results. It does not change calculations.

Read the drivetrain

Battery source feeds the DC-link capacitor, then the inverter sends three-phase power to the permanent-magnet synchronous motor (PMSM). The motor turns a shaft, gear reduction and wheels. The 3D cutaway is generic teaching geometry. Rotor and wheel angles follow the selected calculated shaft angle, with wheel angle divided by the gear ratio. Reduced motion holds the rotating illustration still while readouts and graphs remain usable. No vehicle mass, tyre slip, road grade, distance, battery state of charge or thermal behaviour is calculated.

Read graphs and legends

The vertical scale names the unit and the horizontal scale is virtual seconds. The moving dashed cursor marks the selected instant; cards and the drive view follow it. Solid, dashed and dotted traces distinguish signals alongside colour. Hover, focus or tap an i button by a legend for its meaning. Reference is the requested value; Motor is the calculated response. Compare speed error, response after a load change, overshoot and settling rather than looking at one number alone.

Try and verify one change

Save the baseline CSV first. Choose the changed preset (1000 rpm, lower supply) and wait for completion. Inspect the final speed and compare the time response. Keep other inputs fixed when testing one parameter. Default end speed is about 1500 rpm and DC-link voltage about 298.9 V. Source power, inverter power and shaft power need not match during transients because energy is stored and losses are present.

Understand phases and PWM

Choose the switched PWM preset, then the Phase and PWM waveforms tab. Inspect the final 2 or 20 milliseconds. Three phase currents, phase voltages, phase duties, triangular carrier and all six gate commands are available at solver resolution. In averaged mode, gate traces are illustrative comparisons; in switched mode, they are applied to the bridge. Ideal paired switches have no dead time. Fast rotation and switching may appear to strobe; slow playback or pause to inspect samples.

Independent converter experiments

Buck reduces DC voltage; boost increases DC voltage; the rectifier changes AC into filtered DC. These experiments have separate sources and resistive loads. They are not attached to the motor drivetrain and are not a battery charger. Their inputs are grouped under advanced settings. Defaults settle near 300 V for buck and 400 V for boost. The lossy steady-state preset starts near 292.683 V and 382.979 V respectively.

Export and verify evidence

Recorded CSV covers the run at the selected recording interval. Fine waveform CSV covers the final 20 ms at the solver interval. Both preserve all 51 named columns. The baseline exports 5001 recorded samples plus the header. The switched preset exports 1201 recorded and 10001 fine samples plus headers. Coarse recording can miss switching peaks; use fine data for electrical waveform analysis. Exports are local learning evidence, not a hardware measurement.

Advanced solver and model boundaries

The preserved reference engine has 13 states integrated with fixed-step fourth-order Runge-Kutta. Averaged mode uses 20 microseconds. Switched mode uses the smaller of 2 microseconds and 1/(50 x PWM frequency). Validation rejects settings that exceed the browser step budget, produce unsuitable recording intervals or are too stiff for this fixed solver. Do not try to fix a rejected run by changing the playback rate.

Reference evidence

The supplied Edward programmer guide, implementation handover and portable executable provide the model specification. The preserved numerical suite contains 32 checks and 170 independent embedded MATLAB checkpoints across three scenarios. Original SLX and complete native binary traces were not supplied in this workspace, so the historical comparison of 1,100,003 samples has not been repeated. The 3D assembly improves presentation and does not establish manufacturer geometry, OEM calibration or additional physical validation.

Commands and run - all inputs

Target speed (rpm)

The motor speed you want, in revolutions per minute. The controller tries to reach this value; the measured motor speed is shown separately.

Default: 1500. Signal: speed_rpm.

Speed ramp (s)

How long the target takes to rise from zero to the chosen speed. For example, 0.5 means the target rises over half a virtual second.

Default: 0.5. Signal: ramp_s.

Initial load (N m)

The twisting resistance on the shaft before the load changes. A larger value makes the motor work harder.

Default: 10. Signal: load_before.

Final load (N m)

The twisting resistance after the selected load-change time. Compare it with the initial load to make a harder or easier test.

Default: 20. Signal: load_after.

Load change time (s)

The moment when the shaft load changes. A value of 2 means the load changes two virtual seconds after the run starts. This is NOT the calculation interval.

Default: 2. Signal: step_s.

Run duration (s)

How many virtual seconds to calculate. The opening run lasts 5 seconds. This is experiment time, not the time your computer needs to finish.

Default: 5. Signal: duration.

Recording interval (s)

How often the run saves a row of results. A value of 0.001 saves one row every millisecond. The solver calculates between these saved rows.

Default: 0.001. Signal: record_s.

Battery and DC link - all inputs

Source voltage (V)

The battery source voltage feeding the drive. This model does not calculate how full the battery is.

Default: 300. Signal: Vb.

Feed resistance (ohm)

Resistance between the battery source and DC link. Current through it causes a voltage drop.

Default: 0.1. Signal: Rfeed.

Discharge limit (A)

The maximum current the battery source may supply to the DC link.

Default: 50. Signal: Idis.

Charge limit (A)

The maximum current the source may accept back from the link. This setting does not add a vehicle regenerative-braking experiment.

Default: 0. Signal: Ichg.

DC-link capacitance (F)

The capacitor on the DC link. It temporarily stores electrical energy and helps smooth the inverter supply.

Default: 0.002. Signal: Cdc.

Motor and wheel - all inputs

Stator resistance (ohm)

Resistance in each motor winding. It contributes to electrical losses.

Default: 0.05. Signal: Rs.

d-axis inductance (H)

Inductance on the motor d axis, which describes one direction of current in the rotating motor reference frame.

Default: 0.0005. Signal: Ld.

q-axis inductance (H)

Inductance on the motor q axis. q-axis current is the main torque-producing current in this experiment.

Default: 0.0005. Signal: Lq.

Magnet flux (Wb)

The strength of the permanent-magnet flux used in the motor equations.

Default: 0.08. Signal: psi.

Pole pairs

Number of north-south magnet pairs. Electrical angle turns this many times for one mechanical shaft turn.

Default: 4. Signal: pp.

Shaft inertia (kg m^2)

How difficult it is to speed up or slow down the shaft. A larger inertia gives a slower mechanical response.

Default: 0.05. Signal: J.

Viscous friction (N m s/rad)

A friction term that grows with shaft speed. It makes the motor need extra torque.

Default: 0.001. Signal: B.

Gear ratio

Motor turns per wheel turn. At a ratio of 8, eight motor turns give one wheel turn. This is a simple kinematic conversion.

Default: 8. Signal: gear.

Wheel radius (m)

Wheel radius used to turn wheel rotation into an equivalent speed. No road motion or tyre slip is calculated.

Default: 0.3. Signal: rw.

Controller and PWM - all inputs

Bridge mode

Averaged mode uses smooth bridge duties. Switched PWM applies the individual gate commands and needs much smaller calculation intervals.

Default: 0. Signal: switching.

PWM frequency (Hz)

How many PWM switching cycles occur each second. This is electrical switching frequency, not animation speed.

Default: 10000. Signal: fpwm.

Modulation limit

The allowed modulation size. It limits how much voltage the controller can request from the inverter.

Default: 0.95. Signal: mmax.

q-current limit (A)

Limit on the torque-producing q-axis current request.

Default: 80. Signal: Imax.

Current bandwidth (Hz)

How quickly the current controller is tuned to react. Higher is faster, but must remain compatible with the fixed solver interval.

Default: 500. Signal: fci.

Speed bandwidth (Hz)

How quickly the speed controller is tuned to react. Use small, justified changes when studying response.

Default: 5. Signal: fcs.

Speed damping ratio

The speed controller damping setting. It affects how smoothly the motor settles after a change.

Default: 0.9. Signal: zeta.

Buck experiment - all inputs

Input voltage (V)

The DC input voltage for this independent converter. It is separate from the motor drive source.

Default: 400. Signal: buckVin.

Duty ratio

The fraction of each switching cycle commanded on. 0.75 means 75%. This converter uses averaged equations.

Default: 0.75. Signal: buckD.

Inductance (H)

The converter inductor stores energy in its magnetic field and affects current change.

Default: 0.002. Signal: buckL.

Capacitance (F)

The output capacitor stores energy and helps smooth the output voltage.

Default: 0.002. Signal: buckC.

Load resistance (ohm)

The output load resistance. A smaller resistance draws more current at the same voltage.

Default: 80. Signal: buckR.

Inductor resistance (ohm)

Resistance in the inductor. It represents a loss and changes the steady output.

Default: 0. Signal: buckrL.

Initial condition

Empty startup begins with no stored inductor/capacitor energy. Steady state begins at the calculated equilibrium.

Default: 1. Signal: buckStart.

Boost experiment - all inputs

Input voltage (V)

The DC input voltage for this independent converter. It is separate from the motor drive source.

Default: 300. Signal: boostVin.

Duty ratio

The fraction of each switching cycle commanded on. 0.75 means 75%. This converter uses averaged equations.

Default: 0.25. Signal: boostD.

Inductance (H)

The converter inductor stores energy in its magnetic field and affects current change.

Default: 0.002. Signal: boostL.

Capacitance (F)

The output capacitor stores energy and helps smooth the output voltage.

Default: 0.002. Signal: boostC.

Load resistance (ohm)

The output load resistance. A smaller resistance draws more current at the same voltage.

Default: 80. Signal: boostR.

Inductor resistance (ohm)

Resistance in the inductor. It represents a loss and changes the steady output.

Default: 0. Signal: boostrL.

Initial condition

Empty startup begins with no stored inductor/capacitor energy. Steady state begins at the calculated equilibrium.

Default: 1. Signal: boostStart.

Rectifier experiment - all inputs

AC source RMS (V)

The independent AC source RMS voltage for the rectifier experiment. It is not the drive battery voltage.

Default: 230. Signal: Vac.

AC frequency (Hz)

How many AC cycles the rectifier source produces each second.

Default: 50. Signal: fac.

Rectifier load (ohm)

The independent rectifier load resistance. A smaller value asks for more load current.

Default: 40. Signal: Rload.

Filter capacitance (F)

The rectifier filter capacitor, which helps smooth the pulsating DC voltage.

Default: 0.001. Signal: Cf.

AC source resistance (ohm)

Resistance in the independent rectifier source path.

Default: 0.5. Signal: Rsource.

Drop per diode (V)

Voltage lost across each conducting diode in the rectifier bridge.

Default: 0. Signal: Vd.

Graph and legend glossary

Reference

The target value the controller is trying to follow.

Motor

The calculated motor response. In torque charts this is produced motor torque; in speed charts it is actual speed.

Load

The external twisting resistance on the motor shaft, or resistive load current in a converter chart.

DC link

Voltage on the capacitor feeding the inverter.

Supply into link

Electrical power entering the DC link from the source.

Inverter AC

Electrical power at the ideal inverter AC output.

Electromagnetic

Power converted between electrical and rotating motor states.

Shaft load

Mechanical power absorbed by the imposed shaft load.

Phase A

Calculated value in motor phase A.

Phase B

Calculated value in motor phase B.

Phase C

Calculated value in motor phase C.

Carrier

The triangular comparison signal used to form PWM gate commands.

q reference

The torque-producing current requested by the controller.

q current

Calculated torque-producing current in the rotating reference frame.

d current

Calculated current along the magnet-flux reference axis.

Input

Independent converter input voltage.

Output

Calculated converter output voltage.

Ideal reference

Lossless voltage reference; losses and transients can make output differ.

AC source

The independent alternating voltage entering the rectifier.

Full-wave bridge

The rectified voltage before the output capacitor smooths it.

Filtered DC

Voltage after the rectifier filter capacitor.

Inductor

Current through the converter inductor.

Diode charging

Current charging the filter capacitor through the diode bridge.

Duty A

The requested on-fraction for phase A. 0 means off and 1 means fully on.

Upper A

Ideal upper gate command for phase A: 1 on, 0 off. Paired upper/lower commands are complementary, without dead time.

Lower A

Ideal lower gate command for phase A: 1 on, 0 off. Paired upper/lower commands are complementary, without dead time.

Duty B

The requested on-fraction for phase B. 0 means off and 1 means fully on.

Upper B

Ideal upper gate command for phase B: 1 on, 0 off. Paired upper/lower commands are complementary, without dead time.

Lower B

Ideal lower gate command for phase B: 1 on, 0 off. Paired upper/lower commands are complementary, without dead time.

Duty C

The requested on-fraction for phase C. 0 means off and 1 means fully on.

Upper C

Ideal upper gate command for phase C: 1 on, 0 off. Paired upper/lower commands are complementary, without dead time.

Lower C

Ideal lower gate command for phase C: 1 on, 0 off. Paired upper/lower commands are complementary, without dead time.

Complete 51-column CSV map

time_s

Selected virtual time in seconds.

iq_reference_A

Requested q-axis current, amperes.

vd_command_V

Controller d-axis voltage request, volts.

vq_command_V

Controller q-axis voltage request, volts.

carrier

PWM triangular carrier, 0 to 1.

drive_speed_rpm_1

Requested speed, rpm.

drive_speed_rpm_2

Calculated motor speed, rpm.

drive_torque_Nm_1

Produced motor torque, N m.

drive_torque_Nm_2

External shaft load torque, N m.

dc_bus_V_A_1

DC-link voltage, V.

dc_bus_V_A_2

Source current into the link, A.

drive_power_W_1

Source power into the link, W.

drive_power_W_2

Inverter AC electrical power, W.

drive_power_W_3

Electromagnetic power, W.

drive_power_W_4

Shaft load power, W.

phase_currents_A_1

Phase A current, A.

phase_currents_A_2

Phase B current, A.

phase_currents_A_3

Phase C current, A.

wheel_speed_kmh_1

Equivalent wheel speed from gearing and wheel radius, km/h; no road dynamics.

motor_states_1

d-axis current, A.

motor_states_2

q-axis current, A.

motor_states_3

Mechanical angular speed, rad/s.

motor_states_4

Mechanical shaft angle, radians.

phase_voltages_V_1

Phase A-to-neutral voltage, V.

phase_voltages_V_2

Phase B-to-neutral voltage, V.

phase_voltages_V_3

Phase C-to-neutral voltage, V.

inverter_current_A_1

Inverter DC input current, A.

six_gates_1

Upper A gate.

six_gates_2

Upper B gate.

six_gates_3

Upper C gate.

six_gates_4

Lower A gate.

six_gates_5

Lower B gate.

six_gates_6

Lower C gate.

phase_duties_1

Phase A duty fraction.

phase_duties_2

Phase B duty fraction.

phase_duties_3

Phase C duty fraction.

buck_voltages_1

Input voltage, V.

buck_voltages_2

Output voltage, V.

buck_voltages_3

Ideal reference voltage, V.

boost_voltages_1

Input voltage, V.

boost_voltages_2

Output voltage, V.

boost_voltages_3

Ideal reference voltage, V.

rectifier_voltages_1

AC source voltage, V.

rectifier_voltages_2

Full-wave bridge voltage, V.

rectifier_voltages_3

Filtered DC output, V.

buck_currents_1

Inductor current, A.

buck_currents_2

Load current, A.

boost_currents_1

Inductor current, A.

boost_currents_2

Load current, A.

rectifier_currents_1

Diode charging current, A.

rectifier_currents_2

Load current, A.